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That's a very common finding, and there are several well-documented reasons a manually-computed bulk-Ri PBLH from ERA5 137-level profiles tends to run higher than the archived **blh** field, especially over land:

1. The official diagnostic isn't a "naive" bulk Ri — it has a built-in convective correction

ECMWF's internal PBLH diagnostic (following Vogelesang and Holtslag, 1996, building on Troen and Mahrt, 1986) doesn't just compute

$$Ri_b(z) = g \cdot z \cdot (\theta_v(z) - \theta_{v,surf}) / (\theta_{v0} \cdot (u^2 + v^2))$$

using the raw lowest-level θ_v as the surface reference. In unstable (convective) conditions it adds:

- A **surface-layer gustiness/turbulent velocity term** to the denominator (something like $b \cdot u^{*2}$ added to $u^2 + v^2$) so Ri doesn't blow up to unrealistically low values when actual wind shear is weak — very common over land at midday when winds near the surface are light and buoyancy-driven turbulence dominates.
- In many implementations, a **temperature excess** added to the surface parcel's virtual potential temperature to represent thermals, rather than using the bare lowest-model-level value.

If your own script uses the plain two-term bulk Ri without these correction terms, the behavior you get can go either direction depending on which effect dominates at a given profile — but the gustiness term is usually the big one over land: without it, weak near-surface winds make the denominator artificially small, and Ri can occasionally shoot past 0.25 lower in the profile in one layer, then not cross again until much higher up if there's a secondary near-surface stable layer (like a shallow superadiabatic layer right at the surface before the well-mixed layer proper) — so a naive calculation is prone to picking the *wrong* crossing point, often a higher one, especially if you're not careful about taking the *first* level from the surface where Ri exceeds critical, or if you're interpolating across large model-level spacing.

2. Level-crossing interpolation

With 137 levels, spacing near the surface over land is only a few tens of meters, but by the time you're a km up, spacing can be 200–400 m. If your code just steps level-by-level and

doesn't linearly interpolate Ri between the last sub-critical and first super-critical level, you can overshoot by a full model layer, inflating PBLH by hundreds of meters at coarser resolution aloft.

3. Reference level mismatch

The official diagnostic typically uses the **lowest model level** (not 2 m temperature/humidity) as the surface reference. If you're using 2 m T/Td or skin temperature instead, especially over land at midday when there's a large gradient in the first few meters, you'll compute a different (often larger) $\Delta\theta_v$ aloft needed to cross the threshold.

4. Forecast step vs analysis mismatch

The archived ERA5 **blh** is a **short-range forecast diagnostic** (from the 06/18 UTC forecast cycle), computed at the same instant as the analysis but not literally derived by ECMWF from the archived pressure/model-level fields you downloaded — it's produced online inside the model at full internal resolution and physics coupling. If you're recomputing Ri offline from the archived profile (which has been vertically discretized, possibly interpolated to pressure levels rather than kept on native model levels, and represents an average grid-box state), you're not exactly reproducing what the model's internal diagnostic saw at the physics timestep.

5. Land-specific noise

Land boundary layers, especially convective ones, tend to have noisier near-surface gradients (patchy surface heating, orography, soil moisture heterogeneity) than ocean columns. This makes a purely local Ri profile bounce around near threshold, so which crossing you pick (first exceedance vs. a more robust running criterion) can matter a lot — and the official scheme's extra terms exist partly to suppress that noise.

Practical suggestion: if you want your offline calculation to match ECMWF's number, use native model levels (not pressure-interpolated), the lowest model level as your surface point, linear interpolation to find the exact height where $Ri = 0.25$, and add a wind-gustiness term (something like $+1$ to 2 (m/s)^2 equivalent, or a scaled u^{*2} term) to the shear denominator rather than raw $(u^2 + v^2)$ alone. Comparing your PBLH against the official one on a scatterplot, stratified by stability (stable/neutral/convective) and by land cover, is a good diagnostic — mismatches concentrated in weak-wind convective daytime land cases would confirm it's the gustiness-term issue rather than a bug elsewhere in your code.

for the gustiness $b * usqr$, is there any dependency on surface temperature

Synthesized domain knowledge while acknowledging unverified internal model parameters

Not directly in the term's algebraic form, but yes indirectly, through how u^* itself is diagnosed.

The term itself: In the Vogelezang & Holtslag (1996) formulation, the denominator of the bulk Ri is

$$(u(z) - u_s)^2 + (v(z) - v_s)^2 + b \cdot u^{*2}$$

The $b \cdot u^{*2}$ addition doesn't contain temperature explicitly — it's just a fixed multiple (b is an empirical constant, order 100 in the original Cabauw-tower fit) of the square of the surface friction velocity. So there's no separate T-dependent term bolted on next to it.

But u^* itself is stability-dependent, and stability is set by the surface-to-air temperature difference:

u^* comes out of the surface-layer similarity solution (Monin-Obukhov / Louis-type scheme in IFS), where

$$u^* = \kappa \cdot U(z_1) / [\ln(z_1/z_0) - \psi_m(z_1/L)]$$

and the Obukhov length L is

$$L = -u^{*3} \cdot \theta_v / (\kappa \cdot g \cdot (w' \theta_v')_s)$$

$(w' \theta_v')_s$, the surface buoyancy flux, is driven by the skin-temperature minus near-surface air-temperature difference (plus surface humidity flux). So:

- Over land at midday, strong surface heating (skin T well above the lowest-level air T) → large negative L → unstable ψ_m correction increases effective u^* somewhat relative to the neutral log-law value for the same wind speed.
- At night, skin T below air T → stable, u^* is reduced by the stability correction for the same wind speed.

So the chain is: surface (skin) temperature → sensible heat flux/buoyancy flux → Obukhov length → stability correction to the log-wind profile → u^* . The $b \cdot u^{*2}$ gustiness term inherits that temperature dependence secondhand, through u^* , not because temperature appears explicitly in the $b \cdot u^{*2}$ expression.

One more wrinkle worth flagging: some non-local schemes in this family (Troen-Mahrt 1986 type, used e.g. in YSU/KPP) instead build the unresolved-shear term from a *convective* velocity scale w^* rather than (or in addition to) u^* , and w^* has a much more direct temperature dependence:

$$w^* = [(g/\theta_v) \cdot z_i \cdot (w' \theta_v')_s]^{1/3}$$

Since w^* is a cube root of the surface buoyancy flux times a height scale, if the scheme you're replicating uses w^{*2} in place of (or added to) u^{*2} in the shear term, then yes, the temperature dependence becomes explicit and fairly strong (via the sensible heat flux). Whether ECMWF's operational implementation uses u^* -only or blends in a w^* -based term matters a lot for how sensitive your reconstruction is to skin-air temperature difference — it would be worth checking which exact velocity-scale term is coded in the current IFS Chapter 3 documentation (turbulent transport) rather than assuming it's identical to the original 1996 Cabauw fit, since ECMWF has tuned/modified the original formulation over successive cycles.

OK, so $b=100$ is a constant. But I need an estimate for u^* . I don't have any of these buoyancy terms. So can you distill all this into a simple form for me to use? I'll have skin temperature, u_{10}, v_{10} and that is it. I don't even have T_{10} or T_2

Weighed neutral drag coefficient approach against stability limitations

Fair enough — with only skin temperature and (u_{10}, v_{10}) , you can't close the buoyancy/stability loop properly (that needs at least one temperature *difference*, and skin temperature alone doesn't give you one). Here's the honest, distilled version of what you can and can't do:

What you can compute: a neutral estimate of u^*

$$u_* = \frac{\kappa U_{10}}{\ln(10/z_0)}$$

where:

- $U_{10} = \sqrt{u_{10}^2 + v_{10}^2}$
- $\kappa = 0.4$
- z_0 = roughness length. For land you need this from somewhere — either ERA5's own forecast surface roughness field (`fsr`), a standard single-level ERA5 variable, essentially free to add to your download), or a fixed land-cover-based lookup (~0.03–0.1 m for open land, ~0.1–0.5 m for cropland/scattered trees, ~1+ m for forest/urban).

Then use $b \cdot u^{*2}$ with $b \approx 100$ as before.

What this misses: it's a *neutral* u^* , with no stability correction. The real u^* should be boosted somewhat in unstable conditions (convective, land daytime — exactly your problem case) and reduced in stable conditions, via Monin–Obukhov $\psi_m(z/L)$. Since you have no air temperature at all, you can't form the surface buoyancy flux needed for L , so you literally cannot apply that correction — there's no way around this with the three variables you listed.

Practical implication of skipping it: the neutral u^* will *underestimate* the true u^* in unstable/convective conditions. That makes your gustiness term ($b \cdot u^2$) *too small, which makes the denominator in your Ri formula too small, which makes Ri cross the critical threshold too low* — i.e., *this bias pushes your PBLH estimate down in exactly the convective land cases where you were seeing it come out too high. So it may partially offset the original problem, or it may not — it depends on which effect (missing gustiness term vs. neutral- u underestimate of gustiness) dominates at a given point, and there's no clean way to predict that without testing.*

My actual recommendation: grab `t2m` (2 m temperature) alongside skin temperature — it's a standard ERA5 single-level field, adds essentially no extra download cost, and gives you the one piece you're missing. With T_s (skin) and T_2 you can form a proper bulk Richardson number for the surface layer itself and get a stability-corrected u^* using the Louis (1979/1982) scheme — the same analytic (non-iterative) formulation ECMWF's own surface layer uses — rather than guessing with a neutral approximation. If you want, I can write out that Louis-scheme formula in a form you can drop straight into code once you have T_2 .

Yes can you write down that Lous scheme. Do I use over land and ocean, or on;y over land

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Here's the Louis (1979) surface-layer scheme in the compact, non-iterative form — this is the same conceptual scheme (with ECMWF's slightly retuned constants) that underlies IFS's own surface layer.

Step 1 — Bulk Richardson number

Since z is small, you can skip potential-temperature correction and just use temperature directly:

$$Rib = \frac{g z (T_2 - T_s)}{\bar{T} U^2}$$

- $z \approx 10$ m (use the wind reference height; T_2 and U_{10} aren't at the same level, which is a known approximation you're forced into here)
- $U^2 = u_{10}^2 + v_{10}^2$
- $\bar{T} \approx$ mean of T_2 and T_s (or just use T_2)
- T_s = skin temperature, T_2 = 2 m temperature
- Sign check: $T_s > T_2 \rightarrow$ unstable (surface warmer than air) $\rightarrow Rib < 0$. $T_s < T_2 \rightarrow$ stable $\rightarrow Rib > 0$.

Step 2 — Neutral drag coefficient

$$C_{DN} = \left(\frac{\kappa}{\ln(z/z_0)} \right)^2, \quad \kappa = 0.4$$

z_0 = roughness length (land: use ERA5's `fsr` field or a land-cover lookup table; ocean: see note below).

Step 3 — Stability correction (Louis 1979, $b = c = d = 5$)

Unstable ($Rib < 0$):

$$C_D = C_{DN} \left[1 - \frac{2b Ri_b}{1 + 3bc C_{DN} \sqrt{(z/z_0) |Ri_b|}} \right]$$

Stable ($Ri_b \geq 0$):

$$C_D = \frac{C_{DN}}{1 + 2b Ri_b / \sqrt{1 + d Ri_b}}$$

Step 4 — Friction velocity

$$u_* = \sqrt{C_D} U$$

Then plug that u_* into the $b \cdot u_*^2$ gustiness term from before.

Land vs. ocean: the scheme itself (the $Ri_b \rightarrow$ stability-correction math above) is generic and applies over both surfaces — it's not land-specific. What differs is z_0 :

- **Land:** z_0 is a static/vegetation-dependent field — pull ERA5's `fsr` (forecast surface roughness) directly, or use a land-cover table. Straightforward, no extra work.
- ****Ocean:**** z_0 is *not* static — it depends on the wind stress itself via the Charnock relation:

$$z_0 = \alpha_{ch} \frac{u_*^2}{g} + \frac{0.11\nu}{u_*}$$

($\alpha_{ch} \approx 0.018$, ν = kinematic viscosity of air). This makes z_0 and u_* mutually dependent, so strictly you'd need 2-3 iterations (guess u_* , compute z_0 , recompute u_* , repeat — converges fast). If you don't want to iterate, ERA5 also archives an ocean-specific roughness field you can substitute directly instead of deriving it from Charnock.

So: same formula, but expect an extra iteration step over ocean that you don't need over land. If your problem is specifically about land PBLH discrepancies (as it sounded like from your first question), you can just use the static z_0 approach and skip the ocean complication entirely.

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Didn't really make a difference. If I make that b larger (eg 1000 instead of 100) then that brings the PBLH too large.

The data set I am looking at, ERA5 has PBLH that is lower than I get with the bulk Richardson crossing $R_{critical} = 0.25$

Diagnosed fundamental formula flaw rather than tuning problem

That symptom — bigger b making things worse, not better — actually tells you the gustiness term was never your main problem. You're tuning the denominator, but the bug is almost certainly in the **numerator**: what you're using as the surface $\theta_{v,s}$ reference.

The likely core issue: if you're using skin temperature T_s as $\theta_{v,s}$ in the Rib formula itself (not just for the Louis stability correction, but as the actual "surface" point the profile is compared against), that's a fundamentally different quantity from what the scheme wants. Skin temperature over land during the day can run 10–20°C+ warmer than the near-surface air. That means:

$$Rib(z) = \frac{g z (\theta_v(z) - \theta_{v,skin})}{\theta_{v0} U^2}$$

stays strongly negative for a long stretch of the profile — the actual air is always cooler than the blazing-hot skin, all the way up through the real mixed layer and often into the free troposphere above it. The numerator only turns positive (and starts approaching +0.25) very high up, once you finally reach air cold enough that even a hot skin reference is exceeded appropriately by the linear z -scaling. That gives you a systematically, sometimes dramatically, too-high PBLH — and no amount of retuning b fixes it, because the problem isn't the denominator's magnitude, it's that the numerator crosses zero at the wrong place entirely.

This is consistent with everything you've reported: your height is too high, and pushing the denominator around (bigger b) only makes it worse because a bigger denominator delays the crossing even further, compounding the same underlying bug rather than fixing it.

The fix: the surface reference in Rib must be a near-surface *air* temperature (2 m, or lowest model level), not skin temperature. Skin temperature belongs in the buoyancy-flux/stability calculation ($T_s - T_2$ drives the Obukhov length and hence u^* 's stability correction), but it should not itself be the $\theta_{v,s}$ that goes into the Rib numerator.

Given that, I don't think there's a way to distill this further without T_2 (or the lowest model-level air temperature) — it's not an optional refinement, it's load-bearing for the numerator, not just the gustiness correction. Is there a reason T_2/T_{10} isn't available to you (e.g., a specific dataset/subset that only carries skin temperature and 10 m wind)? If you're pulling from the ERA5 single-level archive, `2m_temperature` is a standard variable right alongside `2m_dewpoint_temperature`, `10m_u_component_of_wind`, etc. — same request, no extra dataset needed. If there's a hard constraint keeping you from adding it, let me know what it is and I can think about whether there's a workaround, but I'd be honest that any workaround will be a much cruder approximation than just grabbing T_2 .

No I do have T_2

Good — that resolves it then. Let me give you the corrected, complete recipe using what you actually have (Ts, T2, u10, v10, plus the profile).

Rib numerator — use T2, not Ts:

$$Rib(z) = \frac{g z (\theta_v(z) - \theta_{v,2m})}{\theta_{v0} [(u(z) - u_{10})^2 + (v(z) - v_{10})^2 + b u_*^2]}$$

- $\theta_v(z)$ computed from your profile levels (T, q, p at each level)
- $\theta_{v,2m}$ computed from T2 (and 2 m specific/dewpoint humidity if you have it — if not, just use T2 as a temperature proxy for $\theta_{v,2m}$, accepting a small error from ignoring moisture)
- z measured from the surface (or from 2 m, being consistent)
- u10, v10 as the reference wind (matches the V&H formulation, which differences against near-surface wind, not zero)

Ts's job is separate — only for stability/u*:

Use Ts and T2 only to get Rib,surface (the Louis-scheme input) → stability correction → u*. That u* then only enters the b·u*² gustiness term in the denominator above. Ts should not appear anywhere in the main numerator.

Then:

- b = 100 (leave it there — you were right to not need to inflate it; the earlier symptom was a numerator problem, not a denominator one)
- z0 from fsr (land) for both the Louis calculation and the neutral CDN
- Critical Ri = 0.25, find the level (interpolating between bracketing levels) where Rib first crosses upward through 0.25 from below, scanning from the surface up

This should now behave properly: the previously "too high" bias driven by comparing air aloft against a hot skin temperature goes away, since T2 tracks the actual near-surface air rather than the radiometric surface skin value.

One thing worth checking once you rerun it: verify sign conventions and units are consistent (θ_v in K throughout, $g=9.81$, z in m) and that you're not double-subtracting the reference level twice (once in z, once implicitly if your profile's first level is 2 m — decide whether z is measured from 0 or from 2 m and be consistent between the wind and temperature terms). If it's still off after this fix, the next place I'd look is the u/v reference — some implementations difference against the *lowest model level* wind rather than u10/v10, and 10 m vs. lowest-model-level winds can differ by a couple m/s in stable conditions, which matters more than you'd think in the shear term.

better! not perfect but better. I will *not* have t2m for my retrieval, since I use a radiativeTransferAlgorithm that is coarser than the ERA5 137 levels.

should I just interpolate the profile (which is probably spaced at 125 m to 250 m resolution

Weighed interpolation pitfalls against using lowest available level directly

Don't interpolate — extrapolating a coarse profile down to 2 m will actively hurt you, not help, and here's why: the layer between your lowest usable level (~125–250 m) and the surface is exactly where the sharpest, most non-linear structure lives — the Monin-Obukhov constant-flux layer, where T follows a log-linear (not linear) profile in stable conditions, and where a strong nocturnal inversion or a shallow daytime superadiabatic layer can sit *entirely* below your resolution. Linearly (or spline-) interpolating your coarse levels down through that layer implicitly assumes the free-atmosphere lapse rate holds all the way to the ground, which is almost never true in the surface layer. You'd be manufacturing a fake T_2 that's often further from reality than just using your lowest level directly.

Better approach: use your lowest available level as the reference, not 2 m.

Everything from before still applies — just substitute your lowest profile level (call it z_1 , T_1 , u_1 , v_1) everywhere 2 m/10 m appeared:

- Surface Rib for the Louis stability correction: use (T_s , T_{z1}) and z_1 instead of (T_s , T_2) and 2 m.
- Rib numerator for the PBLH scan: reference $\theta_v(z_1)$ instead of $\theta_v, 2m$.
- Wind reference in the shear denominator: u_1 , v_1 instead of u_{10} , v_{10} .

This is conceptually identical to what NWP surface layer schemes already do — ECMWF's own lowest model level (~10 m) plays this same role, you're just doing it at coarser height. The math doesn't care what z_1 is, as long as you're consistent.

The caveat you can't engineer around: Monin-Obukhov similarity is only strictly valid within the surface layer, roughly the bottom ~10% of the boundary layer depth. If z_1 is 125–250 m and the actual PBL is deep (convective, >1500 m), you're probably fine — z_1 is still within the constant-flux layer. But for shallow or stable PBLs (nocturnal land, often only 100–300 m deep), your lowest level can end up *at or above* the top of the surface layer, or even above the PBL top itself. In that regime, the whole basis for deriving u^* from (T_s , T_{z1}) via MO similarity breaks down — there's no flux-layer relationship to exploit anymore, because z_1 isn't really "in" the layer the theory describes.

There's no clean fix for that last case with the resolution you have — it's a genuine information limit, not a formula bug. Practically:

- Expect your worst residual errors to cluster in stable, shallow-PBL, low-wind land cases (nighttime), since that's where z_1 most often exceeds the surface layer depth.
- Unstable/convective daytime cases should improve a lot, since the surface layer is typically deeper there and z_1 stays within it.

- If you want a rough self-check, compare your diagnosed PBLH to z_1 itself — if you're finding $\text{PBLH} \lesssim z_1$ or only marginally above it, that's a signal you're in the regime where the surface-layer assumption is shaky, and the number should be treated with more skepticism.

